

Empirical Validation of the Ghoshian Dual Parallel State–Attention Architecture

Soumadeep Ghosh with ChatGPT (OpenAI)

Kolkata, India

Abstract

The Ghoshian Dual Parallel State–Attention Architecture (G-DPSA) was proposed as a synthesis of Transformer reasoning, State Space Model (SSM) memory, and Ghoshian Learning-based executive control. While the theoretical architecture unifies adaptive optimization, dynamic compute allocation, hierarchical memory, reinforcement learning, and mixture-of-experts routing, its practical value depends upon empirical validation.

This paper develops a rigorous empirical framework for evaluating G-DPSA. Rather than immediately attempting large-scale deployment, we identify and test the foundational hypothesis underlying the architecture: that Breadth and Depth constitute measurable state variables of learning dynamics with predictive and potentially causal influence on future learning. We derive econometric, statistical, information-theoretic, dynamical-systems, and machine-learning methodologies for evaluating this hypothesis across Transformers, State Space Models, Vision Transformers, Convolutional Neural Networks, and hybrid architectures.

The resulting framework transforms Ghoshian Learning from a learned optimizer into a candidate theory of optimization dynamics. We propose the Breadth–Depth Learning Atlas, the Breadth–Depth Efficient Frontier, learning phase diagrams, universality tests, scaling-law investigations, and causal intervention experiments. The paper establishes the empirical foundation required for future deployment of Breadth–Depth-controlled G-DPSA systems.

The paper ends with “The End”

Contents

1	Introduction	2
2	Research Questions	3
2.1	Question 1	3
2.2	Question 2	3
2.3	Question 3	3
2.4	Question 4	3
2.5	Question 5	3
3	Theoretical Foundations	4
3.1	Breadth	4
3.1.1	Transformer	4
3.1.2	State Space Model	4
3.1.3	CNN	4
3.2	Depth	4
4	Hypotheses	5
4.1	H1	5
4.2	H2	5
4.3	H3	5
4.4	H4	5
4.5	H5	5

5	Experimental Framework	5
5.1	Future Learning	5
5.2	Classical Model	5
5.3	Ghoshian Model	5
5.4	Statistical Test	5
6	Benchmark Suite	6
6.1	Vision	6
6.2	Language	6
6.3	Reasoning	6
7	Breadth–Depth Dataset	6
8	Information-Theoretic Analysis	7
9	Orthogonalization Tests	7
10	Causal Intervention Framework	7
10.1	Breadth Shock	7
10.2	Depth Shock	7
11	Dynamical Systems Analysis	8
12	Learning Phase Diagram	8
13	Learning Frontier	8
14	Breadth–Depth Efficient Frontier	8
15	Scaling Laws	9
16	Universality Tests	9
17	Economic Interpretation	9
18	Psychological Interpretation	10
19	Political and Institutional Interpretation	10
20	Welfare Economics	10
21	Warfare Economics	10
22	Expected Outcomes	11
22.1	Outcome A	11
22.2	Outcome B	11
22.3	Outcome C	11
23	Conclusion	11

List of Figures

1 Introduction

Modern machine learning has achieved remarkable success through increasingly large Transformer architectures. Simultaneously, State Space Models have emerged as computationally efficient alternatives capable of processing extremely long contexts. Despite these advances, optimization remains governed primarily by externally imposed hyperparameters.

The Ghoshian Dual Parallel State–Attention Architecture was introduced to address this limitation through the integration of:

1. Transformer reasoning,
2. State Space memory,
3. Ghoshian Learning,
4. Reinforcement-learning-based executive control,
5. Adaptive memory allocation,
6. Dynamic attention allocation,
7. Mixture-of-experts specialization.

The architecture is governed by two central quantities:

$$B_t$$

(Breadth)
and

$$D_t$$

(Depth).

Originally introduced as optimization signals, these variables may possess a deeper interpretation as latent state variables of learning dynamics.

The purpose of this paper is to empirically investigate this possibility.

2 Research Questions

The empirical program is organized around five questions.

2.1 Question 1

Is Breadth measurable?

2.2 Question 2

Is Depth measurable?

2.3 Question 3

Do Breadth and Depth predict future learning?

2.4 Question 4

Do Breadth and Depth possess causal influence?

2.5 Question 5

Are Breadth and Depth universal across architectures?

3 Theoretical Foundations

3.1 Breadth

For a general architecture, define

$$B_t = \frac{\text{Var}(Z_t)}{\|E[Z_t]\|^2 + \epsilon}$$

where

$$Z_t$$

depends on architecture.

Examples include:

3.1.1 Transformer

$$Z_t = A_t$$

where

$$A_t$$

denotes attention maps.

3.1.2 State Space Model

$$Z_t = s_t$$

where

$$s_t$$

denotes hidden states.

3.1.3 CNN

$$Z_t = g_t$$

where

$$g_t$$

denotes gradients.

Breadth measures exploratory diversity.

3.2 Depth

Define

$$D_t = \exp\left(-\gamma \frac{\Delta \tilde{L}_t}{\tilde{L}_{t-1} + \epsilon}\right)$$

where

$$\tilde{L}_t = \alpha \tilde{L}_{t-1} + (1 - \alpha)L_t.$$

Depth measures accumulated convergence.

4 Hypotheses

4.1 H1

Breadth predicts future learning.

$$I(B_t; Y_t) > 0.$$

4.2 H2

Depth predicts future learning.

$$I(D_t; Y_t) > 0.$$

4.3 H3

Breadth and Depth jointly outperform classical optimization variables.

4.4 H4

Breadth and Depth possess causal influence.

4.5 H5

Breadth and Depth are approximately universal.

5 Experimental Framework

5.1 Future Learning

Define future learning gain:

$$Y_t^{(k)} = L_t - L_{t+k}$$

for

$$k \in \{1, 5, 10, 50, 100\}.$$

5.2 Classical Model

Baseline:

$$Y_t = \alpha + \beta_1 L_t + \beta_2 \|g_t\| + u_t.$$

5.3 Ghoshian Model

Extended:

$$Y_t = \alpha + \beta_1 L_t + \beta_2 \|g_t\| + \beta_3 B_t + \beta_4 D_t + u_t.$$

5.4 Statistical Test

Null:

$$H_0 : \beta_3 = \beta_4 = 0.$$

Alternative:

$$H_A : (\beta_3, \beta_4) \neq (0, 0).$$

6 Benchmark Suite

6.1 Vision

Dataset	Model
MNIST	MLP
CIFAR10	CNN
CIFAR10	ResNet18
CIFAR100	ViT-Tiny

Table 1: Vision benchmarks.

6.2 Language

Dataset	Model
TinyStories	Transformer
FineWeb Subset	Transformer
FineWeb Subset	SSM

Table 2: Language benchmarks.

6.3 Reasoning

Dataset	Benchmark
GSM8K	Mathematical Reasoning
MATH	Advanced Mathematics
HumanEval	Code Generation
MMLU	General Knowledge

Table 3: Reasoning benchmarks.

7 Breadth–Depth Dataset

Every optimization step generates

$$X_t = (L_t, \|g_t\|, B_t, D_t, \eta_t, A_t, M_t, C_t).$$

Stored variables include:

```
step
epoch
loss
accuracy
gradient_norm
breadth
depth
learning_rate
momentum
memory_usage
flops
latency
future_loss_reduction
```

The resulting dataset forms the Breadth–Depth Learning Atlas.

8 Information-Theoretic Analysis

Mutual information is estimated using k-nearest-neighbor estimators.

We measure

$$I(B_t; Y_t)$$

$$I(D_t; Y_t)$$

$$I(B_t, D_t; Y_t).$$

The key quantity is

$$\frac{I(B_t, D_t; Y_t)}{I(L_t, \|g_t\|; Y_t)}.$$

Values greater than one imply additional information.

9 Orthogonalization Tests

To determine whether Breadth and Depth merely proxy for existing variables, define

$$B_t^\perp$$

and

$$D_t^\perp$$

as residuals after regressing on

$$L_t, \|g_t\|, \eta_t.$$

We then estimate

$$Y_t = \alpha + \gamma_1 B_t^\perp + \gamma_2 D_t^\perp + u_t.$$

Significant coefficients imply genuinely new information.

10 Causal Intervention Framework

Prediction alone is insufficient.

We require intervention.

10.1 Breadth Shock

Inject gradient noise:

$$g'_t = g_t + \xi_t.$$

Measure resulting changes in

$$Y_t.$$

10.2 Depth Shock

Perturb

$$\gamma$$

in the depth metric.

Measure resulting convergence behavior.

11 Dynamical Systems Analysis

Training trajectories become

$$(B_t, D_t).$$

Define

$$\frac{dB}{dt} = f(B, D)$$

and

$$\frac{dD}{dt} = g(B, D).$$

Optimization becomes a dynamical system.

12 Learning Phase Diagram

We hypothesize four regimes.

Regime	Breadth	Depth
Initialization	Low	Low
Exploration	High	Low
Discovery	High	High
Convergence	Low	High

Table 4: Proposed learning phases.

These regimes can be identified via clustering.

13 Learning Frontier

Define expected future learning:

$$F(B, D) = E[Y|B, D].$$

The learning frontier is

$$(B^*, D^*) = \arg \max F(B, D).$$

This represents the most productive optimization state.

14 Breadth–Depth Efficient Frontier

Inspired by portfolio theory.

Expected gain:

$$\mu(B, D) = E[Y].$$

Risk:

$$\sigma(B, D) = \sqrt{Var(Y)}.$$

Optimal states maximize gain for a given level of uncertainty.

15 Scaling Laws

Let

$$N$$

denote model size.
Estimate

$$\bar{B}(N) = aN^b$$

and

$$\bar{D}(N) = cN^d.$$

Potentially yielding optimization scaling laws.

16 Universality Tests

For architecture

$$A$$

define

$$F_A(B, D).$$

The universality hypothesis is

$$F_A(B, D) = F(B, D)$$

for all architectures.

If supported, Breadth and Depth become universal coordinates of learning.

17 Economic Interpretation

Breadth resembles volatility and diversification.

Depth resembles compounding and capital accumulation.

Economics	Learning
Volatility	Breadth
Compounding	Depth
Portfolio Allocation	Compute Allocation
Capital	Parameters
Central Bank	Ghoshian Controller

Table 5: Economic interpretation.

Optimization becomes a computational economy.

18 Psychological Interpretation

Psychology	Architecture
Working Memory	Transformer
Long-Term Memory	SSM
Executive Function	Controller
Exploration	Breadth
Expertise	Mixture of Experts
Metacognition	Reinforcement Learning Layer

Table 6: Psychological interpretation.

19 Political and Institutional Interpretation

The architecture resembles a federal system.

Institution	Module
Ministries	Experts
Civil Service	SSM
Legislature	Routing
Executive	Controller
Strategic Planning	RL Layer

Table 7: Institutional interpretation.

Breadth represents diversity.

Depth represents institutional maturity.

20 Welfare Economics

Define welfare

$$W = \lambda_Q Q + \lambda_E E - \lambda_C C.$$

where:

- Q = quality,
- E = efficiency,
- C = compute cost.

The controller seeks

$$\max W.$$

21 Warfare Economics

Military organizations balance:

1. reconnaissance,
2. logistics,
3. command,
4. execution.

Analogously:

- Breadth provides reconnaissance,
- SSM provides logistics,
- Transformer provides command,
- Controller provides strategy.

22 Expected Outcomes

22.1 Outcome A

Strong predictive and causal evidence.

Breadth and Depth become new optimization observables.

22.2 Outcome B

Moderate predictive evidence.

Useful diagnostics.

22.3 Outcome C

No evidence.

Theory falsified.

Still scientifically valuable.

23 Conclusion

The empirical validation program transforms the Ghoshian Dual Parallel State–Attention Architecture from a theoretical proposal into a testable scientific framework.

The central contribution is the hypothesis that Breadth and Depth are measurable state variables governing learning dynamics. Through econometric analysis, information theory, causal interventions, dynamical-systems reconstruction, scaling-law analysis, and universality testing, the framework seeks to determine whether optimization possesses a low-dimensional hidden state space.

If supported by empirical evidence, Breadth and Depth would cease to be merely optimization heuristics and instead become candidate macroscopic coordinates of learning itself. Such a result would provide the empirical foundation for future Breadth–Depth-controlled G-DPSA systems and potentially for a broader statistical mechanics of machine learning.

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Glossary

Breadth Measure of exploratory diversity within learning dynamics.

Depth Measure of accumulated convergence and refinement.

G-DPSA Ghoshian Dual Parallel State–Attention Architecture.

Learning Atlas Dataset of optimization trajectories in Breadth–Depth space.

Learning Frontier Optimal region maximizing expected future learning.

Efficient Frontier Risk-return frontier in Breadth–Depth space.

Universality Hypothesis that Breadth and Depth are architecture-independent coordinates.

Exploration Learning regime characterized by high Breadth.

Convergence Learning regime characterized by high Depth.

Ghoshian Controller Executive layer governing optimization and resource allocation.

The End